Denote by $\mathscr{F}_h^{(k)}$ the flow map of the truncated Hamiltonian expansion $\bar{H}_k$,

$$\mathscr{F}_h^{(k)} \stackrel{\text{def}}{=} \exp(h\mathcal{L}_{\bar{H}_k}).$$

By construction, we have

$$z_{n+1} = \mathscr{F}_h^{(k)}(z_n) + \eta_n$$

where $\|\eta_n\| \le Ch^{k+1}$.

Since $\mathscr{F}_h^{(k)}$ preserves its Hamiltonian (3.11), we have

$$\bar{H}_k(\mathscr{F}_h^{(k)}(z_n)) = \tilde{H}_k(z_n).$$

Now

$$\begin{aligned}
\bar{H}_k(z_\nu) - \bar{H}_k(z_0) &= \sum_{n=0}^{\nu-1} \bar{H}_k(z_{n+1}) - \bar{H}_k(z_n) \\
&= \sum_{n=0}^{\nu-1} \bar{H}_k(z_{n+1}) - \tilde{H}_k(\mathscr{F}_h^{(k)}(z_n)) \\
&= \sum_{n=0}^{\nu-1} \bar{H}_k(\mathscr{F}_h^{(k)}(z_n) + \eta_n) - \tilde{H}_k(\mathscr{F}_h^{(k)}(z_n)).
\end{aligned}$$

Hence

$$|\bar{H}_k(z_\nu) - \bar{H}_k(z_0)| \le L \sum_{n=0}^{\nu-1} \|\eta_n\| \le L\nu h^{k+1}.$$

We have $\nu = \tau/h$, thus

$$|\bar{H}_k(z_\nu) - \bar{H}_k(z_0)| \le L\tau h^k.$$

Next observe that

$$H = \bar{H}_k - h^r H_{(r)} - h^{r+1} H_{(r+1)} - \ldots - h^k H_{(k)} = \bar{H}_k + \mathcal{O}(h^r).$$

Therefore

$$|H(z_\nu) - H(z_0)| \le L\nu h^{k+1} + \mathcal{O}(h^r),$$